

The countable powers of the Arens space and the sequential fan are not k_R -spaces

Candidate full answer to Question 5.11 of arXiv:1610.02522

11 August 2026

Status: candidate full result, likely valid. In ZFC,

$$(S_2)^\omega \quad \text{and} \quad (S_\omega)^\omega$$

are both *not* k_R -spaces. The proof embeds the paper's metric test space P_0 as a closed subspace of a tail of either power. The S_ω case then contradicts the strict closed-fan obstruction already proved in the source. The S_2 case descends to the same obstruction through the standard perfect quotient $S_2 \rightarrow S_\omega$.

1 The question and idea of the proof

Lin, Lin, and Liu ask whether the countable powers $(S_2)^\omega$ and $(S_\omega)^\omega$, known not to be k -spaces, might nevertheless be k_R -spaces [1, Question 5.11]. Here S_2 is the Arens space and S_ω is the sequential fan.

The source also introduces the metrizable space

$$P_0 = \{p\} \cup \bigcup_{i \in \mathbb{N}} X_i,$$

where every X_i is countably infinite, discrete, and open, and

$$V_k = \{p\} \cup \bigcup_{i \geq k} X_i \quad (k \in \mathbb{N})$$

is a neighborhood base at p . Its Lemma 3.6 proves

$$S_\omega \times P_0 \text{ is not a } k_R\text{-space.} \tag{1}$$

The missing observation is that P_0 occurs canonically and *closedly* inside a countable product of any space containing a closed countably infinite discrete subspace. Both S_2 and S_ω have such a subspace. Stratifiability then makes the closed embedding useful: closed subspaces of stratifiable k_R -spaces are k_R [2, Prop. 5.11].

2 A universal closed copy of P_0

Give $\mathbb{N}$ the discrete topology and distinguish $0 \in \mathbb{N}$. In Baire space $\mathbb{N}^\omega$, put

$$E = \{0^\omega\} \cup \{e_{i,m} : i \in \omega, m \in \mathbb{N}\}, \quad e_{i,m}(j) = \begin{cases} m, & j = i, \\ 0, & j \neq i. \end{cases}$$

Lemma 1. *The set E is closed in $\mathbb{N}^\omega$ and is homeomorphic to P_0 . Consequently, if a space T contains a closed countably infinite discrete subspace, then T^ω contains a closed copy of P_0 .*

Proof. The complement of E consists exactly of sequences with at least two nonzero coordinates. Specifying those two coordinates gives an open cylinder disjoint from E , so E is closed.

Each $e_{i,m}$ is isolated in E . If $X_i = \{e_{i,m} : m \in \mathbb{N}\}$ and $p = 0^\omega$, then the cylinder fixing the first k coordinates to zero meets E in

$$\{p\} \cup \bigcup_{i \geq k} X_i.$$

These cylinders form a neighborhood base at p , which is precisely the definition of P_0 .

If $D \subset T$ is closed, countably infinite, and discrete, then D^ω is closed in T^ω and is homeomorphic to $\mathbb{N}^\omega$. The displayed $E \subset D^\omega$ is therefore the required closed copy. $\square$

For completeness, the needed discrete sets can be chosen explicitly. Write

$$S_\omega = \{a_0\} \cup \{a(n, m) : n, m \in \mathbb{N}\},$$

where every $a(n, m)$ is isolated and a neighborhood of a_0 contains a tail of every branch. Then

$$D_\omega = \{a(n, 1) : n \in \mathbb{N}\}$$

is closed and discrete: the neighborhood with branch threshold 2 misses all of it. In the usual notation

$$S_2 = \{\infty\} \cup \{x_n : n \in \mathbb{N}\} \cup \{x_n(m) : n, m \in \mathbb{N}\},$$

the set $D_2 = \{x_n(1) : n \in \mathbb{N}\}$ is likewise closed and discrete. Tail neighborhoods of each x_n , followed by the defining neighborhoods of ∞ , separate every nonmember from D_2 .

3 The obstruction $S_\omega \times P_0$

We recall why (1) holds. Enumerate $X_n = \{b(n, m) : m \in \mathbb{N}\}$ and set

$$F_n = \{(a(n, m), b(n, m)) : m \in \mathbb{N}\} \subset S_\omega \times P_0.$$

Every F_n is clopen. Indeed, its points are isolated; and since X_n is clopen in P_0 , a point outside F_n is separated either from X_n in the second coordinate or from the unique paired point in the first coordinate. The family $\{F_n : n \in \mathbb{N}\}$ is compact-finite: the projection of a compact subset of S_ω can meet only finitely many branches away from a_0 . To see the latter fact, choosing one point from each of infinitely many met branches would produce an infinite closed discrete subspace, because one may choose the branch thresholds at a_0 to miss all selected points.

On the other hand, the family is not locally finite at (a_0, p) . Given a neighborhood of this point, choose n beyond the prescribed P_0 -tail and then m beyond the prescribed threshold on the n -th fan branch; the neighborhood meets F_n . Because the F_n are clopen, they form a strict closed fan. A k_R -space contains no strict closed fan [1, Lemma 3.2], proving (1).

4 The negative answer

We use three standard facts, recording their exact role.

1. Stratifiable spaces are also called M_3 -spaces, and countable products of M_3 -spaces are M_3 [3, Thm. 2.4]. The sequential fan is a closed image of a metrizable topological sum of convergent sequences, hence is Lašnev and therefore stratifiable. The Arens space S_2 is a 1-paracomplex and hence stratifiable [4, Thm. 2.2 and introduction].
2. If A is closed in a stratifiable k_R -space X , then A is k_R [2, Prop. 5.11].
3. Quotient images of k_R -spaces are k_R . Indeed, if $q : X \rightarrow Y$ is quotient and $g : Y \rightarrow \mathbb{R}$ is continuous on compact subsets, then $g \circ q$ is continuous on each compact $K \subset X$ because $q(K)$ is compact. Hence $g \circ q$ is continuous, and the quotient property makes g continuous.

Theorem 2. *Neither $(S_\omega)^\omega$ nor $(S_2)^\omega$ is a k_R -space.*

Proof. First suppose that $(S_\omega)^\omega$ is k_R . Split off its first coordinate. By Lemma 1, its remaining coordinates contain a closed copy of P_0 ; hence the whole power contains a closed copy of $S_\omega \times P_0$. The power is stratifiable by Fact 1, so Fact 2 would make this closed subspace k_R , contradicting (1).

Now suppose that $(S_2)^\omega$ is k_R . The same construction gives a closed copy

$$S_2 \times P_0 \subset (S_2)^\omega.$$

Again the ambient power is stratifiable, so $S_2 \times P_0$ is k_R .

Collapse the compact convergent sequence $C_0 = \{\infty\} \cup \{x_n : n \in \mathbb{N}\} \subset S_2$ to one point. The resulting quotient is S_ω . The quotient map $q : S_2 \rightarrow S_\omega$ is perfect: for a closed $F \subset S_2$, its saturation is either F or $F \cup C_0$, so q is closed; its exceptional fiber is the compact set C_0 , and all other fibers are singletons. Perfect maps are stable under products, so

$$q \times \text{id}_{P_0} : S_2 \times P_0 \longrightarrow S_\omega \times P_0$$

is a closed quotient map. Fact 3 would make $S_\omega \times P_0$ a k_R -space, once more contradicting (1). $\square$

5 Audit, scope, and novelty

The proof is entirely in ZFC. It uses no cardinal-characteristic hypothesis, no unverified compactness classification, and no claim that the k_R -property is closed-hereditary in arbitrary spaces. Stratifiability is essential at precisely the two closed-subspace steps. The quotient step is also safe: $q \times \text{id}_{P_0}$ is quotient because it is perfect, not because arbitrary products preserve quotient maps.

The construction answers both parts of Question 5.11, but it does not settle the source's broader Question 5.9 about arbitrary products of sequential spaces. It also does not address the other open questions about free Abelian topological groups.

Exact-title, exact-question, S_2^ω , S_ω^ω , k_R , and citation searches through 11 August 2026 did not locate a published answer to Question 5.11. The closest later items found study unrelated properties of S_2^ω , or establish the stratifiability input for S_2 , but do not state this k_R -conclusion. The core argument is a new combination of the source's own obstruction with the elementary closed P_0 -embedding. Novelty confidence is moderate and independent expert review is requested.

References

- [1] F. Lin, S. Lin, and C. Liu, *The k_R -property of free Abelian topological groups and products of sequential fans*, Topology and its Applications 240 (2018), 78–97; arXiv:1610.02522, <https://arxiv.org/abs/1610.02522>.

- [2] T. Banach and S. Gabrielyan, *On the C_k -stable closure of the class of (separable) metrizable spaces*, Monatshefte für Mathematik 180 (2016), 39–64; arXiv:1412.2216, <https://arxiv.org/abs/1412.2216>.
- [3] J. G. Ceder, *Some generalizations of metric spaces*, Pacific Journal of Mathematics 11 (1961), 105–125, <https://doi.org/10.2140/pjm.1961.11.105>.
- [4] S.-H. Kim, *The Arens space S_2 is a paracomplex: a note on Nagata's Problem 3*, preprint, 3 March 2026, <http://newton.kias.re.kr/~kimsh/papers/2026-nagata-problem-3.pdf>.